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Note

Gait asymmetry detection in older adults using a light ear-worn sensor

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Abstract

Measuring gait asymmetry is an important feature when characterizing functional imbalance between limbs. This could be due to pathologies, such as osteoarthritis, stroke, or associated with the effects of surgeries such as hip arthroplasty. Generally, the study of asymmetry or imbalance has required the use of a gait lab or force plates, which could be expensive and difficult to use in home settings. This work validates the use of a light weight ear sensor (7.4 g) with an instrumented treadmill for 64 subjects (age (60.04 (15.36)) including healthy subjects (14) as well as subjects who had been treated for hip (17), knee-replacement surgery (21) and knee osteoarthritis (12). Subjects performed treadmill walking at several speeds on flat surfaces, inclines and declines. Accelerometer data from the ear sensor were segmented into consecutive steps and temporal features were extracted. The measures of gait cycle time and step-period asymmetry obtained from the ear sensor matched well those of the treadmill for flat surfaces, inclines and declines. The key implication of the study is that the proposed method could replace expensive equipment for monitoring temporal gait features in clinics as well as free-living environments, which is important for monitoring rehabilitation after surgery and the progress of diseases affecting limb imbalance.

Keywords: sensors, gait, wireless, asymmetry

(Some figures may appear in colour only in the online journal)

1. Introduction

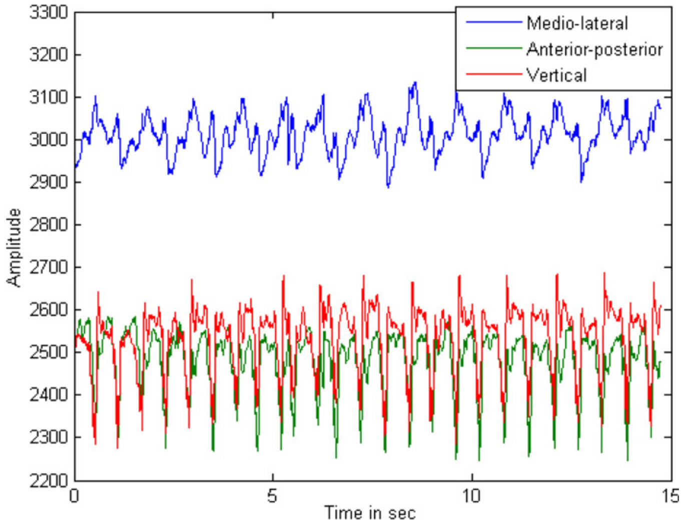
Osteoarthritis is one of the ten most disabling diseases in developed countries. Worldwide estimates are that 10% of men and 18% of women aged over 60 years have symptomatic osteoarthritis, including moderate and severe forms (WHO 2010, Auvinet *et al* 2003). Hip- and knee-replacement surgeries are considered the most effective interventions for severe osteoarthritis, reducing pain and disability and restoring many patients to near-normal function. On average the rate of these surgeries has increased by over 25% in most OECD countries over the last decade, contributing to health expenditure growth. The observation of gait and neuromuscular pattern changes over time allows the longitudinal assessment of the changes in osteoarthritis severity levels, as well as recovery after surgical interventions (Metcalfe *et al* 2013).

Measuring gait asymmetry, connected to several pathologies, is a cardinal feature when characterizing functional imbalance between limbs. This imbalance could be due to the effect of surgery such as hip arthroplasty (Lugade *et al* 2010). Gait asymmetry is also associated with ankle impairments (Lin *et al* 2006), including fractures (Besch *et al* 2008) as well as stroke (Patterson *et al* 2010) and brain lesions. Parameters that have been investigated for the study of asymmetry include the symmetry index, which is a measure of percent differences between sides (Karamanidis *et al* 2003). The symmetry index can be applied to parameters extracted from force plate or treadmill analysis, including peak forces, loading rate, impulse and stance time (McCrory *et al* 2001). The symmetry angle, deduced by measuring the angle formed when a right side value is plotted versus a left side value (Zifchock *et al* 2008), is also used for studying gait asymmetry avoiding the choice of reference values which affects symmetry index values. Generally, the study of asymmetry using these parameters has required the use of a gait lab or force plates, despite the recognized benefit of using pervasive sensing techniques (Kavanagh and Menz 2008). The effect of inclines and declines on asymmetry values also poses a pertinent research question because they may be more demanding for patients whose limb function is deficient.

Quantifying gait parameters that reflect limb strength and asymmetry using inexpensive sensors is a useful tool for observing different groups of subjects in free-living environments without resorting to elaborate gait-labs or treadmill tests. Driven by increasing sensor miniaturization, a number of wearable devices have been deployed to quantify gait parameters. Pressure insoles, for example, have been used mostly in laboratory environments to infer inverse dynamics from motion and observe ground reaction forces (GRF) (Rouhani *et al* 2010, Fong *et al* 2008, Shu *et al* 2010). Accelerometry has recently been applied to an increasing number of studies investigating gait-parameter detection (Kavanagh and Menz 2008). Thus far, the use of accelerometers has been mainly focused on the detection of temporal parameters including walking speeds (Motl *et al* 2012), cadence and step length (Brandes *et al* 2006, Jasiewicz *et al* 2006), heel contact and toe-off events (Moe-Nilssen and Helbostad 2004, Selles *et al* 2005). Force-related features detected by accelerometers, investigated to a lesser extent than temporal features, include sub-plantar GRF (Lo *et al* 2009, Veltink *et al* 2005), weight acceptance peak force, and maximal forces (Atallah *et al* 2012). The quantification of temporal and loading parameters offers indications of atypical joint motion that can be linked to the development of osteoarthritis (Kaufman *et al* 2001). It also provides the basis for assessing recovery after surgery to the knees, legs and hips.

With wearable sensing, appropriate sensor placement is key to the repeatability and consistency of the results and intuitive positions include the trunk (Moe-Nilssen and Helbostad 2004) and legs (Sabatini *et al* 2005). The trunk segment plays an important role in modulating the structure of gait-related oscillations prior to reaching the head during gait. Recent studies

(a)



(b)

Figure 1. Subject wearing the ear worn sensor shown in (a). The 3D accelerometry data from the 3 axes for a subject walking on a treadmill are shown in (b). The right and left steps can be clearly identified in the anterior posterior and vertical axes.

have shown that accelerometers positioned on the head provided smoother oscillations of higher power at low frequencies when assessing walking than those positioned on the trunk (Kavanagh *et al* 2005, 2006). An ear-worn sensor (figure 1(a)) has been previously validated by using an instrumented treadmill to derive gait cycle features as well as weight acceptance peak force and maximal forces (Atallah *et al* 2012). It was also used for predicting energy expenditure (Atallah *et al* 2011) and post-operative monitoring (Aziz *et al* 2007). In this work, the sampling frequency of the ear-mounted sensor was increased from 50 to 130 Hz, allowing the detection of more detailed high-frequency motion features. In addition to that, this work aims to validate the use of this sensor to obtain relevant gait asymmetry features for older adults (64 subjects) who include a normal cohort as well as adults with osteoarthritis and those who have had knee or hip surgery. Based on the literature cited above, it was hypothesized that the values of gait cycle time and gait cycle time asymmetry would be equally detected by the ear-worn sensor and the instrumented treadmill for all subjects who took part in this work.

2. Methods

2.1. Subjects and procedures

64 subjects—34 females and 30 males—age (60.04(15.36) years (mean (SD)), height 169.64(10.58) cm, weight 82.159(15.63) kg, BMI 28.63(4.4) took part in the experiment. Fourteen of the subjects were healthy controls; age 39.7(17), 21 were knee-replacement patients; age 67.9(7.39), 17 were hip-replacement patients; age 65.94(6.49) and 12 were patients who had knee osteoarthritis; age 66.75(10.22). The knee- and hip-replacement patients took part in this experiment 12 months after their operations and furthermore were discharged from clinic follow-up as they were doing very well.

Each subject walked on a force plate instrumented treadmill at speeds starting from 4 km h^{-1} and increasing by 0.5 km h^{-1} until the subject's walking limit was reached (up to 8 km h^{-1}). Subjects then walked at 4 km h^{-1} with increasing inclines of 5% starting from 0 to 20% maximal incline. Participants also walked on a declining slope at 4 km h^{-1}. Before the experiment, all subjects walked for 6 min on the treadmill to acquaint themselves with treadmill walking. Throughout the experiment, subjects wore a miniaturized ($5.6 \times 3.5 \times 1.0\,\text{cm}^3$) and light-weight (7.4 g) ear worn sensor. The sensor housed a 3D accelerometer which measured medio-lateral, vertical and anterior–posterior acceleration at a sampling frequency of 130 Hz. Sensor data was marked by an observer who took note of the speed and incline at each stage. An example of the raw accelerometry data is shown in figure 1(b).

An HP Cosmos Gaitway instrumented treadmill incorporating 2 force plate units housed under the treadmill conveyor belt (gait analysis running machine h/p/cosmos® gaitway II, Germany) was used in this work. The piezoelectric force plates measured vertical GRF which were used by the treadmill software to calculate centre of pressure data and discriminate between right and left footsteps, allowing raw data to be analysed per step. The ear sensor and treadmill data were synchronized by the user pushing the start button on each modality at the same time.

2.2. Analysis framework

Data from the ear-worn sensor was recorded wirelessly and pre-processed using polynomial fitting to subtract the effects of head-motion. We used a data window per treadmill speed and considered the 3 (raw) accelerometry axes separately (shown in figure 1(b)), then fitted a third-order polynomial to the data. This polynomial fit was then subtracted from each axis signal to correct for an overall change in head tilt and position. Note that if there were extreme head motions, this method would not correct for them, as it looks at the overall period rather than local changes. A moving mean filter with a window of three samples was then deployed to smooth the data locally. Given the overall sampling was 130 Hz, this local averaging did not affect the shape or features of the accelerometer signal. A peak detection algorithm and a thresholding step were used to segment the data into steps across the 3 axes. The peak detection consisted of using Matlab algorithms ('peakdet.m') to locate local minima, and then use a threshold to identify the beginning of each step. Points where the minima matched over all 3 axes were selected as overall minima and individual steps were located. After locating each step by using the methods explained above, a measure of step-period asymmetry was obtained by calculating the absolute value of the difference between odd and even step periods per subject from the ear sensor data. Using the absolute value of the difference makes the method robust to the order in which consecutive steps are considered, whether it is left then right or vice versa. Note that we calculated an average for each walking period and matched the two modalities rather than matching step by step data as perfect time synchronization was not possible.

Bland–Altman plots, 95% confidence intervals of the difference between the two means, and standard error were used to analyse agreement between gait cycle times and step-period asymmetry derived by the two methods. Data was analysed using specific code written in Matlab (Mathworks, Inc., Cambridge, UK). Box plots were used to observe the values of gait cycle time as well as gait cycle time asymmetry between the different subject groups.

3. Results

3.1. Predicting gait cycle time

The gait cycle time for the ear sensor was obtained by multiplying the average step time by 2. This value was compared to the values obtained from the treadmill for all speeds, inclines and declines in table 1 (one value per speed and incline/decline rather than step by step matching). The values of mean difference of 0.02 s (2% of the average gait cycle time) show a good match between the two methods in determining gait cycle time regardless of subject. The limits of agreement are around 10% of the average gait cycle time. The patient group is not a homogeneous cohort and with the variation of speed, we do see a variation in the values detected. The Bland–Altman plot presented in figure 2(a) further clarifies the results. Unlike the earlier results presented in Atallah *et al* (2012), the error was not affected by changes of speed although all subjects were free to move their head. The Bland–Altman plots for the individual groups are shown in figures 2(b)–(e). The graphs are generally consistent with similar values for bias and limits of agreement. The Bland–Altman plots (figures 2(a)–(e)) show a pattern of diagonal lines appearing. This is due to the clustering of gait cycle times per treadmill speed. The treadmill speed was varied in increments of 0.5 km h^{-1} and that causes the values of gait cycle time to cluster per speed.

The box plots for the gait cycle time values are also shown for four groups in figure 3(a): control subjects, hip-replacement, knee-replacement and osteoarthritis patients. The figures also show a good match between the values detected by the two modalities per group. Note the similar patterns shown by the means (circles) in the two modalities per group.

3.2. Measuring step-period asymmetry

The step-period difference value (or asymmetry) in seconds was compared to the step-period differences between right and left legs (also in absolute value) from the instrumented treadmill. Table 1 shows a good correspondence between the two methods for detecting step-period asymmetry regardless of incline and walking speed, with a mean difference of 0.01 s. Looking at the results further, the match between the gait asymmetry values from the two modalities is better for inclines (0.001 s), which probably posed more difficulty for the subjects participating in this study. The Bland–Altman plot (figure 2(f)) shows the results further, where the values generally show a better match for low step-period differences. Figure 3(b) shows a box plot

Table 1. This table shows a comparison of values for the gait cycle time and the step-period asymmetry from the ear-worn sensor and the treadmill. The table shows the number of points analysed per axis (N), the mean difference, the SD for the difference between the two values, as well as the lower and upper limits of agreement between the two values, including 95% confidence intervals. The Bland–Altman plot for these data is shown in figure 2(a) for the gait cycle time and figure 2(b) for the step-period asymmetry.

Gait cycle time					
Data	N (number of points)	Mean difference (s)	Standard deviation (SD) (s)	Lower limit of agreement (s)	Upper limit of agreement (s)
All speeds and inclinations	545	0.02 (0.01, 0.02) 2% (of the average gait cycle time)	0.05 5% (of the average gait cycle time)	−0.09 (−0.09, −0.08) 9% (of the average gait cycle time)	0.12 (0.11, 0.13) 12% (of the average gait cycle time)
Step-period asymmetry					
Data	N (number of points)	Mean difference (s)	Standard deviation (SD) (s)	Lower limit of agreement (s)	Upper limit of agreement (s)
All speeds and inclinations	276	−0.01 (−0.01, −0.01)	0.07	−0.16 (−0.18, −0.14)	0.15 (0.13, 0.16)
No incline	181	−0.01 (−0.02, 0)	0.07	−0.16 (−0.17, −0.13)	0.14 (0.12, 0.15)
Only decline	50	−0.01 (−0.02, 0)	0.07	−0.2 (−0.25, −0.15)	0.18 (0.13, 0.23)
Incline only	45	0.001 (−0.02, 0.02)	0.07	−0.13 (−0.16, −0.09)	0.14 (0.1, 0.17)

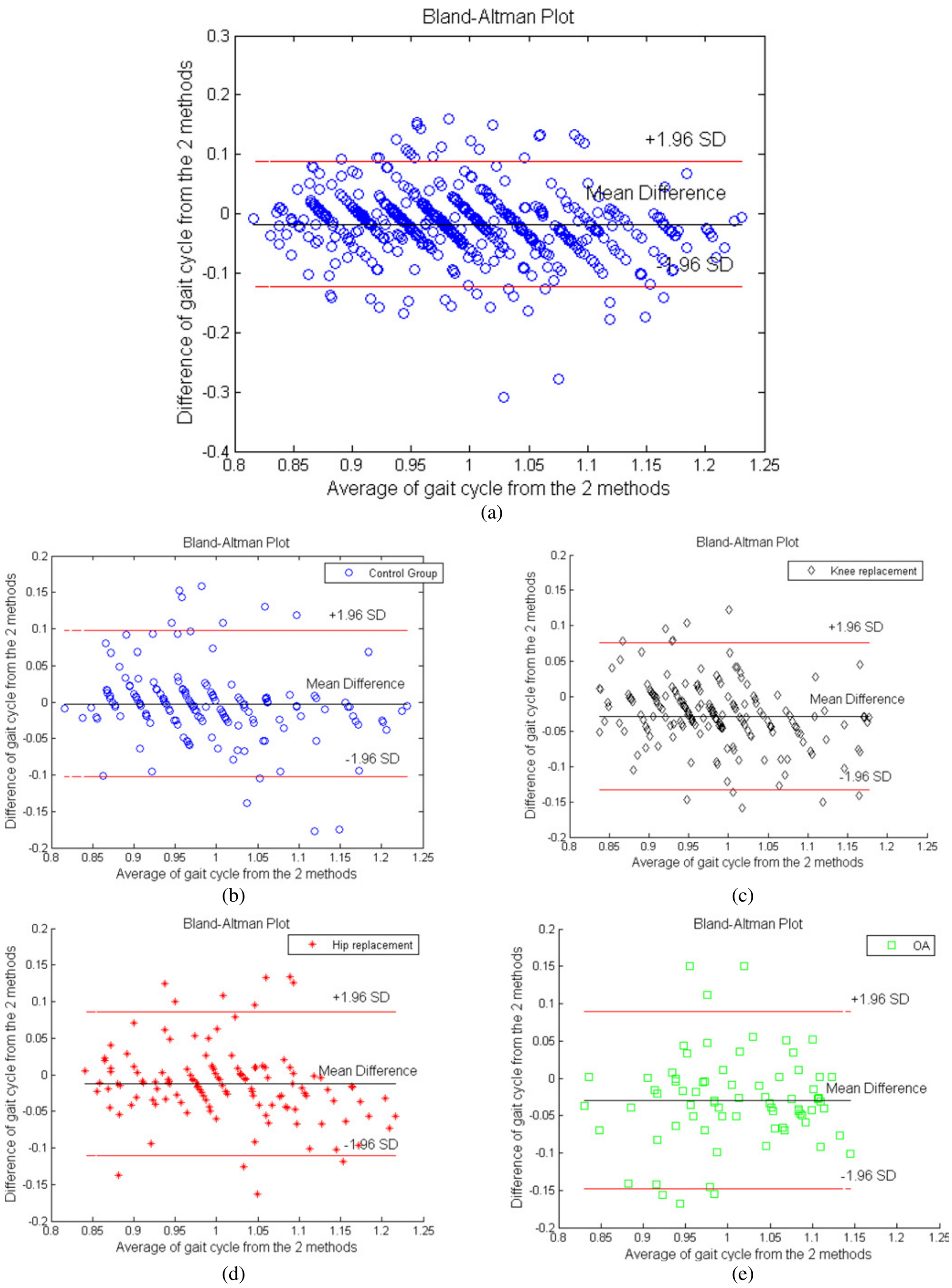
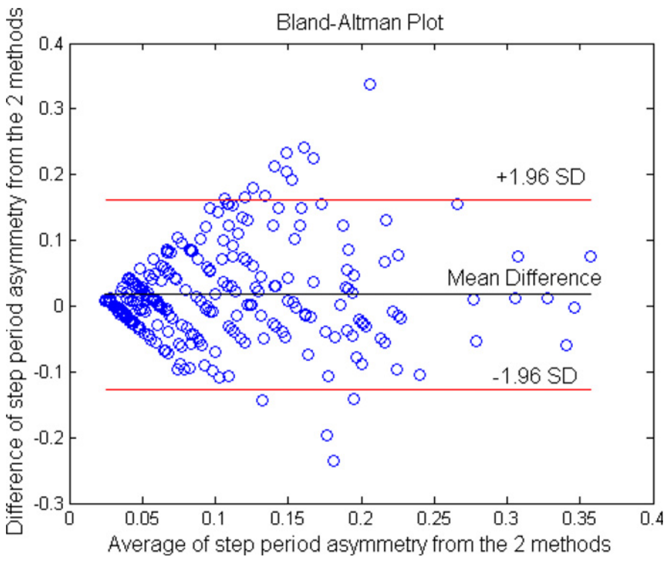


Figure 2. Bland–Altman plots for the gait cycle time from both the e-AR sensor and the treadmill in (a), then for each of the individual groups in (b)–(e) and Bland–Altman plots for step-period asymmetry in (f).



(f)

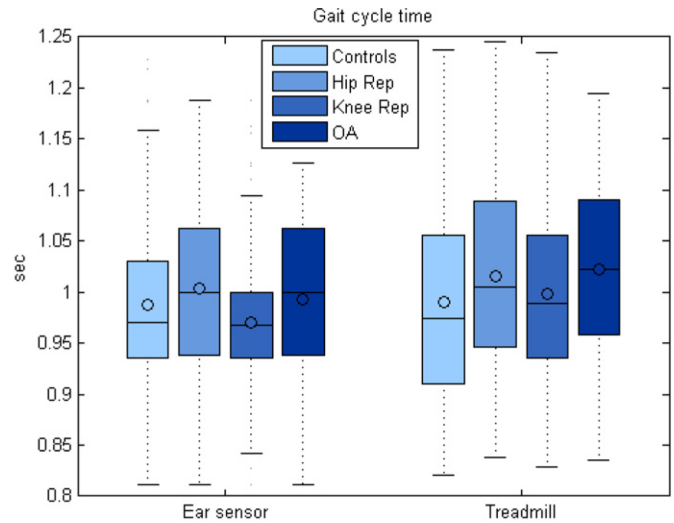
Figure 2. (Continued.)

of the asymmetry values for four groups, which demonstrates a good similarity between the ear sensor and treadmill values per group.

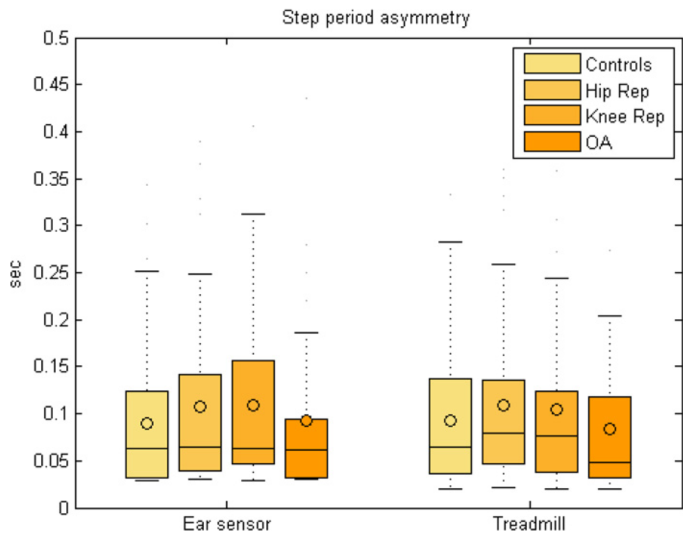
4. Discussion

The results confirmed that a light (7.4 g) ear-worn sensor was capable of determining gait cycle time, and step-period asymmetry regardless of inclines or speed. This compared favourably to the previous approach of calculating zero-crossings per axis (Atallah *et al* 2012). Although the subjects ($N = 64$) were a combined cohort of healthy subjects and patients, the absolute difference between right and left leg period asymmetries were efficiently calculated in order to obtain a measure of asymmetry regardless of which leg was impaired. Since subjects for hip- and knee-replacement participated 12 months after surgery, many of them showed quite symmetric walking. The data from the instrumented treadmill and the ear sensor show similar patterns for these patient groups. Although accelerometers have been used before for studying loading and spatiotemporal gait parameters (Brandes *et al* 2006, Jasiewicz *et al* 2006), the effect of varying inclines on temporal features extracted from head-worn accelerometers has not been investigated previously. The effect of inclines and declines on predicting gait-parameters and asymmetry indicators was investigated in this work (table 1), with inclines showing a better match between the two modalities for gait asymmetry detection, as the mean difference value (0.001) is smaller than the others (0.01).

Furthermore, the ease of use of the sensor, its stable positioning on the head and its consistency in capturing ground reaction forces (Atallah *et al* 2012) as well as temporal features (shown in this study) provide a novel tool for gait assessment. These features make it an attractive option for several applications relating to elderly care. The first one is in the observation of gait features and asymmetry in labs as a means of assessing surgical recovery or the severity levels of osteoarthritis (Astephen *et al* 2008). Compared to the cost of force-plate instrumented treadmills or instrumented gait labs, the use of this sensor is cost effective and can



(a)



(b)

Figure 3. Box-plots of the gait cycle time values and step-period asymmetry values for the four groups: control subjects, hip-replacement, knee-replacement and osteoarthritis patients. The ACL patients were not considered for comparison due to their low number (only six patients were available). (a) shows the gait cycle time values from the ear sensor and the instrumented treadmill, respectively, whereas (b) shows the step-period asymmetry values from the ear sensor and the instrumented treadmill, respectively.

be easily incorporated into rehabilitation programmes or assessment clinics. We have shown in previous work that we could efficiently identify walking periods in free living environments (Atallah *et al* 2011). Combining that with continuous gait cycle time and asymmetry detection would present a novel approach for remote monitoring of sensorimotor control and surgical recovery which is currently not provided by a single visit to gait labs. Changes in gait patterns can indicate a deterioration of balance which could be indicative of a high risk of falling

(Masud and Morris 2001). Falling among adults older than 65 years of age has been reported to occur mainly at home or in the garden, and is the leading cause of injury and death among the elderly (Masud and Morris 2001). A means of observing gait changes remotely could play an important role as more elderly choose to stay at home.

The limitations of this work include the use of fixed speeds and inclines on a treadmill without allowing subjects to walk at their natural pace. Such a controlled environment could lead to less asymmetry and gait cycle time variability. However, the treadmill test (with controlled speeds) is used as a standard to collect gait features, and we just added a lightweight sensor to an existing protocol.

Another limitation of the study is that the control group was younger than the other groups. Since the aim was to capture a symmetrical normal gait and focus on temporal features for controls, we do not feel that their age played an important role in this work. If force-loading, swaying and balance features were to be considered, this could have been an important limitation. The selection of subjects who came back 12 months after surgery as well as those with osteoarthritis served the purpose of obtaining a large dataset for the validation of the ear sensor versus the treadmill. This study was not intended to compare these groups. In fact, figure 3 shows that from a temporal asymmetry perspective, 12 months after hip- and knee-replacement was enough time for these subjects to have temporal asymmetric gait values that are similar to the healthy control group. The same applies to the osteoarthritis patients whose asymmetry values do not show large variations from the control subjects. An extension of this study is to monitor similar subjects, before, just after and on regular periods after surgery.

From a technical point of view, we did not investigate the effect of severe motions, or head rotations on the features obtained as we applied an overall correction that does not take this into consideration. Means of addressing this problem could include automatically detecting changes using temporal models and correcting for such changes. We also did not analyse the variability between individual subjects as we combined several subjects with the same condition to test the overall match between the two modalities.

An important extension would be to test this method in free walking conditions and observe the features shown in this work. An example would be the use of this sensor for elderly home dwellers assessing how the severity and progress of osteoarthritis affect gait monitored using the ear-worn sensor versus standard measures used in clinics. The combination of light weight sensing, such as the sensor presented here, with mobile communications technology means that assessments formerly undertaken in the artificial environment of a gait laboratory can be transferred to a patient's dwelling in a very cost efficient manner.

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